



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025 26)
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Branch	CSE
Course Outcome	CO-2

Case Study : Warehouse Management & Analytical Query System

Work Done Summary:

- Designed a warehouse dataset model with product ID, quantity, and sales logs
- Implemented B-Tree/B+ Tree conceptually for indexing large records
- Implemented Segment Tree for range-based queries (min/max/sum)
- Implemented Fenwick Tree for prefix sum analytics (daily sales tracking)
- Compared performance of each structure based on query type
- Verified time complexity efficiency using sample dataset operations
- Demonstrated correct selection of tree based on use-case (indexing vs analytics)

Implementation Details

1. Product and sales data are logically organized using tree-based indexing for fast retrieval.
2. B-Tree/B+ Tree concept is used for disk-friendly storage with multi-key nodes and balanced height.
3. Segment Tree is implemented for interval queries like total stock or max shipment in a range.
4. Fenwick Tree is used for prefix sum operations such as cumulative daily sales tracking.
5. Update operations propagate efficiently using logarithmic time complexity structures.
6. System ensures optimized performance with $O(\log n)$ query and update operations for all dynamic operations.

Output:

```

int[] bit;
int n;

FenwickTree(int n) {
    this.n = n;
    bit = new int[n + 1];
}

void update(int i, int val) {
    while (i <= n) {
        bit[i] += val;
        i += i & -i;
    }
}

int query(int i) {
    int sum = 0;
    while (i > 0) {
        sum += bit[i];
        i -= i & -i;
    }
    return sum;
}

}

class SegmentTree {
    int[] tree;
    int n;

    SegmentTree(int[] arr) {
        n = arr.length;
        tree = new int[4 * n];
        build(arr, node: 0, l: 0, n - 1);
    }
}

```

```

PS D:\dsa 2 case studies> d:; cd 'd:\dsa 2 case studies'; & 'C:\Program Files\Java\jdk-21\bin\java.
xe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\balap\AppData\Roaming\Code\User\worksp
ceStorage\ea494962941128a6f64542a6540eaa20d\redhat.java\jdt_ws\dsa 2 case studies_68bfcacc\bin' 'Smar
LogixC02'
Fenwick Prefix Sum (first 3 days): 60
Segment Tree Range Sum (1 to 4): 140
PS D:\dsa 2 case studies>

```

The system uses B+ Trees for indexing, Fenwick Tree for prefix sum queries, and Segment Tree for range queries. These data structures ensure efficient $O(\log n)$ operations, improving fast data retrieval, analytics, and overall scalability in warehouse management.

Github link: <https://github.com/yaswanthsurya-99/DSA-2>

Student Signature	Faculty Signature
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